

MODULE - I: Differential Equations

Differential Equation

A differential equation is an equation involving differentials or differential coefficient.

Example:

$$(i) \frac{dy}{dx} = \cot x$$

$$(ii) \frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

$$(iii) y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3$$

Ordinary d. Eqⁿ.

The d. eqⁿ. involving derivatives w.r.to a single independent variable is called an ordinary d. eqⁿ.

Example:

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

Partial d. Eqⁿ.

A d. Eqⁿ. which involve two or more independent variables and partial derivatives w.r.to them is called partial d. Eqⁿ.

Example:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$$

Linear d. Eqⁿ.

A d. Eqⁿ. is linear when the dependent variable and its derivative occur only in first degree and there no any product of the dependent variable and its derivative occur.

Non-linear d. Eqⁿ.

A d. Eqⁿ. which is not linear is called non-linear d. Eqⁿ.

Order

The order of d. Eqⁿ. is the order of highest ordered derivative occurring in the d. Eqⁿ.

Example:

$$\frac{dy}{dx} = \cot x \quad \{\text{Order} = 1\}$$

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0 \quad \{ \text{Order} = 2 \}$$

$$y = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^3 \quad \{ \text{Order} = 1 \}$$

Degree

The degree of d.Eqⁿ is the degree or power of the highest ordered derivative occurring in the equation when it is made free from radical signs and fractional power.

Example:

$$(i) \quad \left(\frac{d^3 y}{dx^3} \right)^4 - 6x^2 \left(\frac{dy}{dx} \right)^8 = 0$$

$$\left\{ \begin{array}{l} \text{Order} = 3 \\ \text{Degree} = 4 \end{array} \right\}$$

$$(ii) \quad \frac{d^2 y}{dx^2} + \sqrt{1 + \left(\frac{dy}{dx} \right)^3} = 0$$

$$\Rightarrow \left(\frac{d^2 y}{dx^2} \right)^2 = 1 + \left(\frac{dy}{dx} \right)^3$$

$$\left\{ \begin{array}{l} \text{Order} = 2 \\ \text{Degree} = 2 \end{array} \right\}$$

Formation of d.Eqⁿ.

Form the d.Eqⁿ by eliminating the arbitrary constant.

(i) $y = Cx^2$

Diff. w.r.to x ,

$$\frac{dy}{dx} = 2Cx$$

$$\Rightarrow C = \frac{1}{2x} \frac{dy}{dx}$$

Substituting,

$$y = \left(\frac{1}{2x} \frac{dy}{dx} \right) x^2$$

$$\Rightarrow 2y = x \frac{dy}{dx}$$

$$\Rightarrow x \frac{dy}{dx} - 2y = 0.$$

$$\left\{ \begin{array}{l} \text{Order} = 1 \\ \text{Degree} = 1 \end{array} \right\}$$

(ii) $y = A \cos 3x + B \sin 3x$

Diff. w.r.to x ,

$$\frac{dy}{dx} = A(-3 \sin 3x) + B 3 \cos 3x$$

$$\text{or, } \frac{dy}{dx} = 3[-A \sin 3x + B \cos 3x]$$

Again diff. w.r.to x ,

$$\frac{d^2y}{dx^2} = 3[-A \cdot 3 \cos 3x + B(-3 \sin 3x)]$$

$$\text{or, } \frac{d^2y}{dx^2} = -9(A \cos 3x + B \sin 3x)$$

$$\text{or, } \frac{d^2y}{dx^2} = -9y$$

$$\Rightarrow \frac{d^2y}{dx^2} + 9y = 0$$

$$\left\{ \begin{array}{l} \text{Order} = 2 \\ \text{Degree} = 1 \end{array} \right\}$$

Linearly Independent & Dependent d.Eqⁿ.

Consider the second order d.Eqⁿ,

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = 0 \quad \text{--- (1)}$$

where a_0, a_1, a_2 are continuous in the interval (a, b) and $a_0 \neq 0$. Then

Then two solution $y_1(x)$ and $y_2(x)$ of equation first are said to be linearly dependent if there exist two arbitrary constant C_1 and C_2 such that,

$$C_1 y_1 + C_2 y_2 = 0 \quad \forall (a, b)$$

The two solution $y_1(x)$ and $y_2(x)$ are said to be linearly independent if they are not to be dependent.

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Linear Differential Equation with Constant Coefficients

A d. Eqⁿ,

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = Q \quad \text{--- (1)}$$

where $a_0 \neq 0$ and $a_1, a_2, \dots, a_n$ are constant and Q is a function of x (or may be constant).

The general / complete solution of eqⁿ is,

$$y = CF \quad [\text{when } Q = 0]$$

$$\& y = CF + PI \quad [\text{when } Q \neq 0]$$

Complimentary Function

Particular Integral

The given d. eqⁿ (1) can be written as,

$$a_0 D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \dots + a_n y = Q$$

$$[a_0 D^n + a_1 D^{n-1} + \dots + a_n] y = Q$$

$$\boxed{f(D). y = Q} \quad \text{--- (2)}$$

$$\text{where, } D^n = \frac{d^n}{dx^n}, \quad D = \frac{d}{dx}$$